

The problem: Agents limited by training data

Introduction

From agent fundamentals to tools

You've learned the core agent equation:

None

`Agent = model + tools + orchestration`

This course completes the core agent equation by teaching you to equip agents with tools—the capabilities that transform agents from intelligent responders into action-capable assistants.

The problem:

Agents limited by training data

Let's revisit an agent from course 3:

```
Python
# From Course 3 - sophisticated agent without tools
from google.adk.agents import LlmAgent

agent = LlmAgent(
    model='gemini-2.5-flash',
    instruction="""
    You are a helpful assistant that provides users with current information.
    Always be accurate and up-to-date in your responses.
    """,
)
```

Now consider these user requests

User: “What's the weather in Tokyo right now?”

User: “Calculate the shipping cost for a
5kg package to Canada”

User: “Look up order #ORD12345 in our system”

The root problem

The agent is limited to what the LLM knows from its training data. For current information, external data, or real-world actions, you need tools.

The fundamental limitations

- ❌ **LLM only knows training data:**
Information is frozen at training time, cannot access real-time data
- ❌ **Cannot perform calculations:**
While LLMs can reason about math, they can't reliably execute precise computations
- ❌ **Cannot access external systems:**
No ability to query databases, call APIs, or interact with services
- ❌ **Cannot take actions:**
Can only generate text, cannot perform operations in the real world